

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU
PRACTICAL - 11

Aim: Implementing Advanced Visualization Techniques and Customizations.

- a) Create slope charts and lollipop charts
- b) Design waterfall charts
- c) Build sparklines for trend analysis
- d) Implement sheet swapping for dynamic dashboards
- e) Create custom geographic territories
- f) Use background images in dashboards
- g) Apply animation and transparency techniques

A: Create Slope Charts and Lollipop Charts

Task 1: Create a Slope Chart (Year-over-Year Progression)

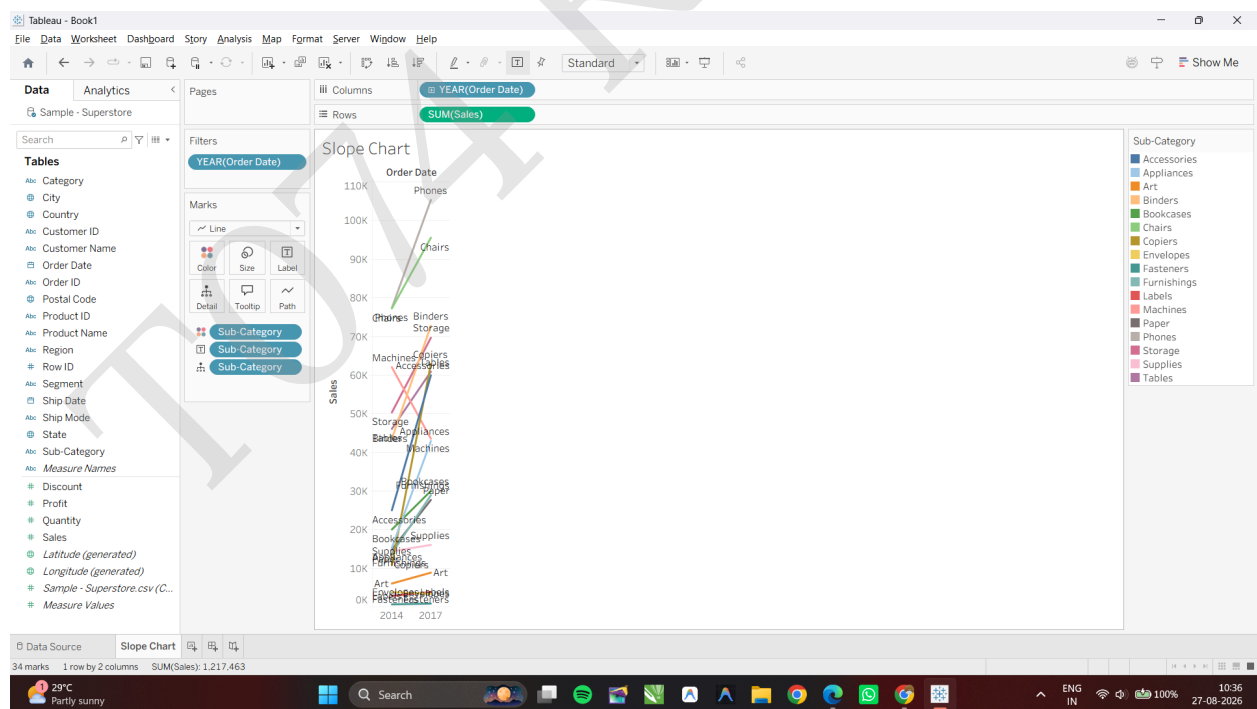
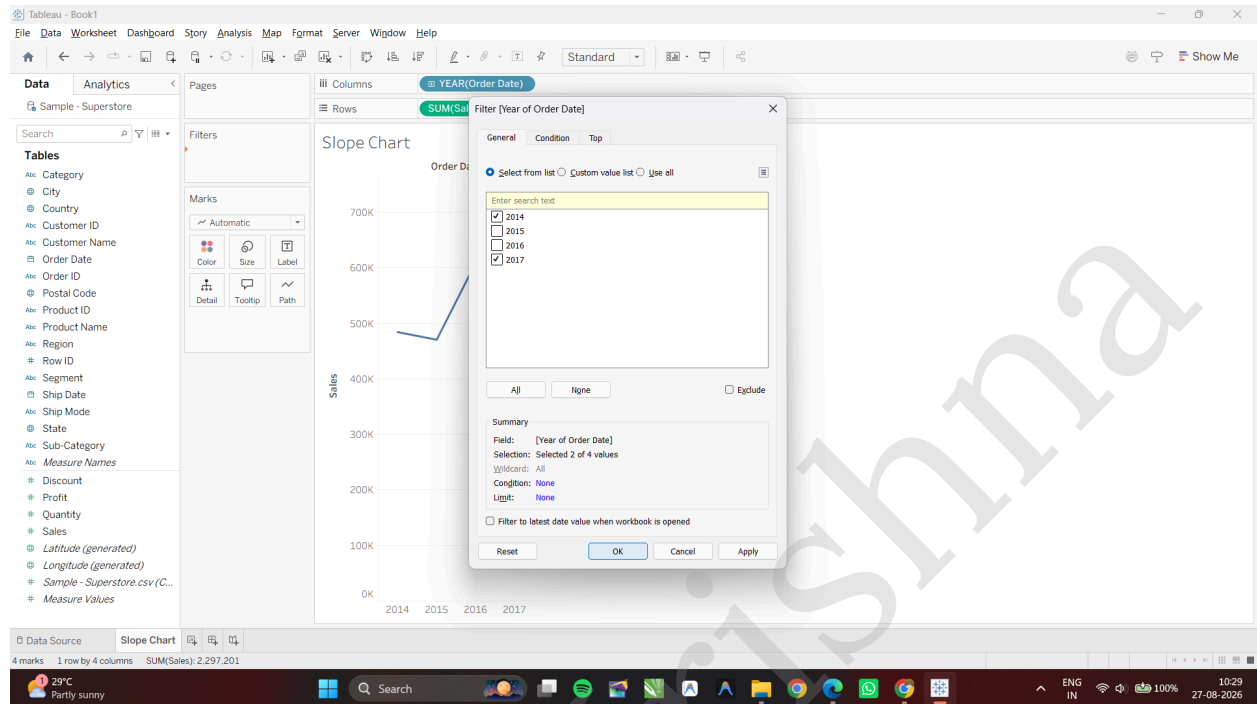
Create a **Slope Chart** by plotting **SUM(Sales)** for **2014 and 2017** using **YEAR(Order Date)**, adding **Sub-Category** to Detail and Color, then enabling labels for both ends of each line.

The screenshot shows the Tableau Desktop interface with the 'Sample - Superstore' data source loaded. The 'Fields' pane on the left shows the data source structure. The main view displays a data table with the following columns and rows:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID
1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520
2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520
3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045
4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335
5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335
6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710

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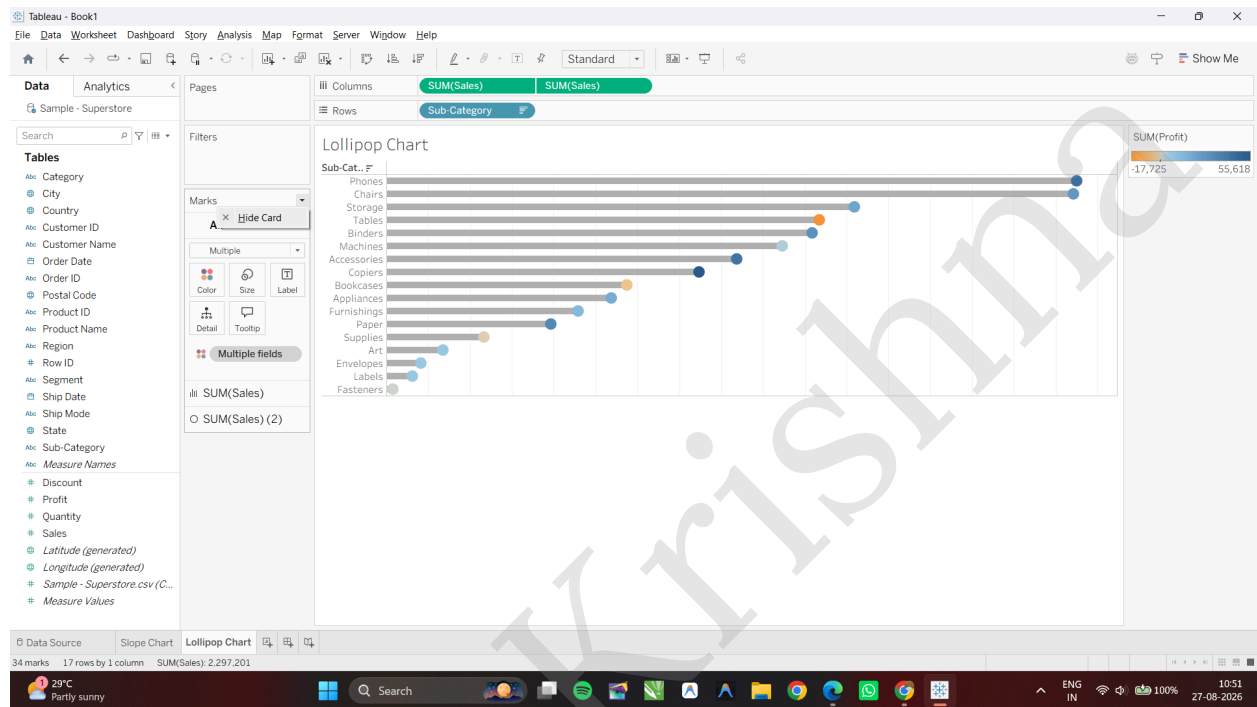


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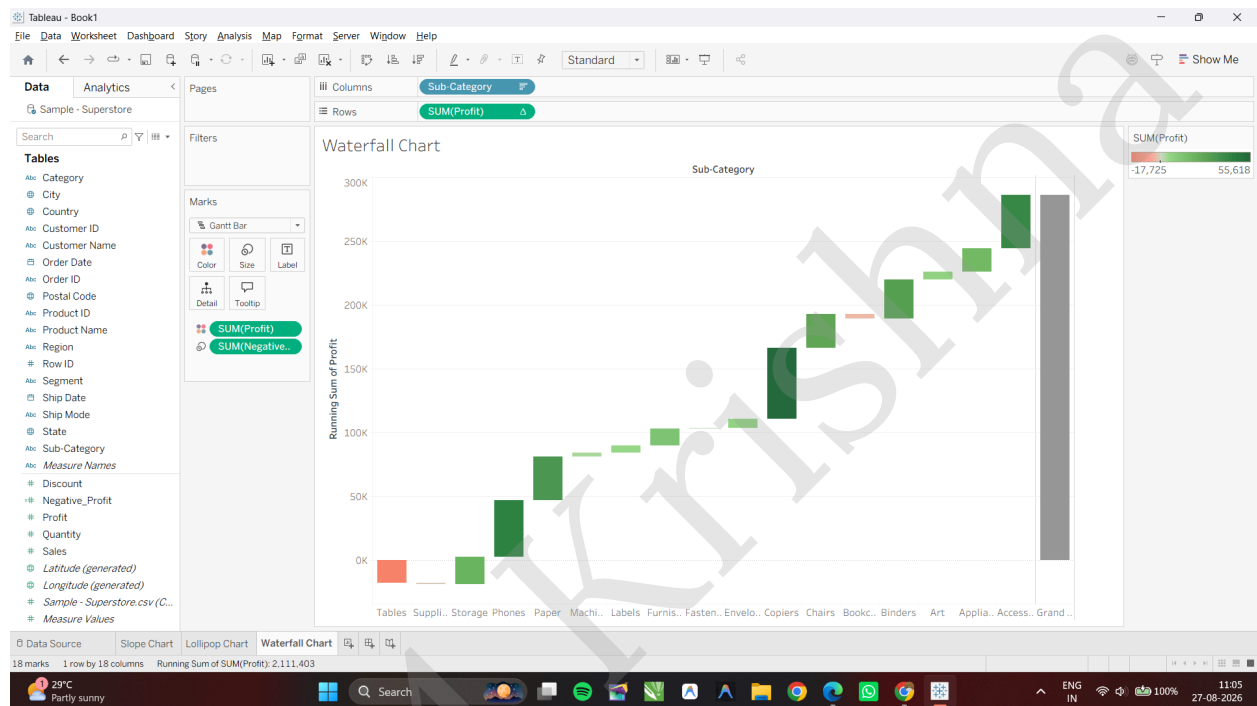
Task 2: Create a Lollipop Chart (Sales with Profit Highlighting)

Create a **Lollipop Chart** by placing **Sub-Category** on Rows and **Sales twice** on Columns, converting them to a **Dual Axis**, using a thin **Bar** for the stem and a **Circle colored by Profit** for the head, then hiding the extra header and sorting by Sales.



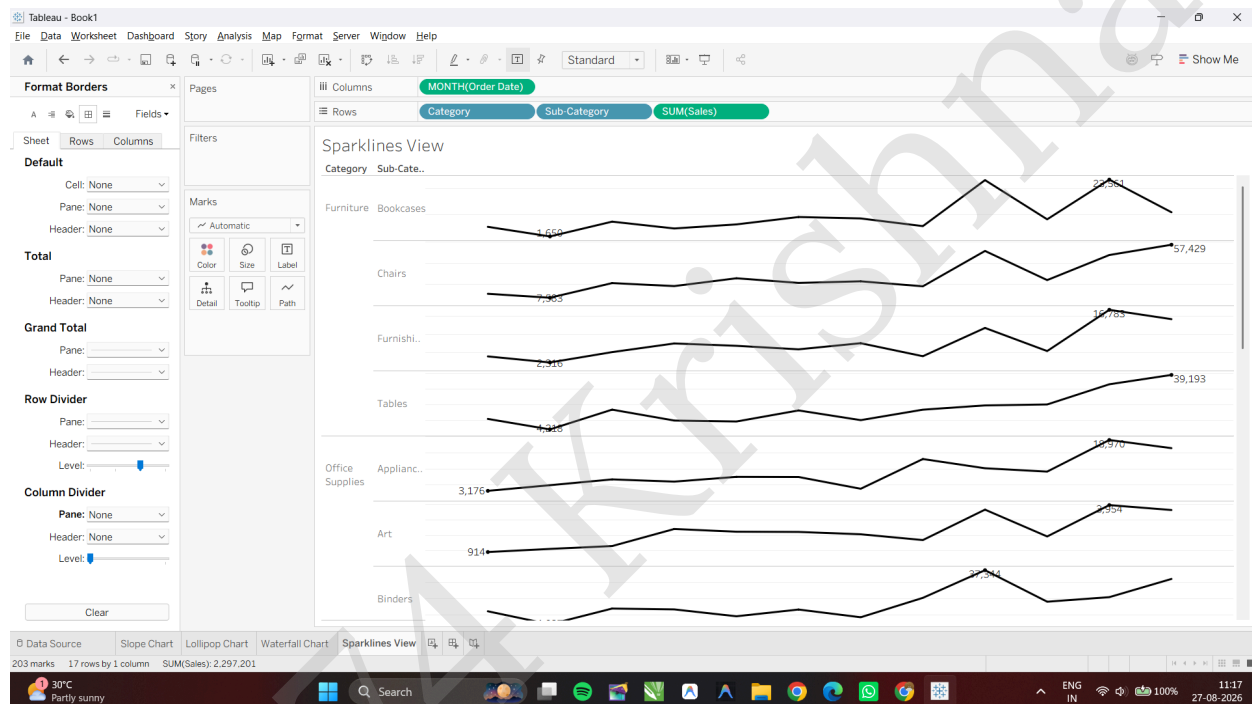
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B: Design Waterfall Charts

Create a **Waterfall Chart** by plotting **Sub-Category** vs. **Profit**, applying a **Running Total**, changing the mark type to **Gantt Bar**, using **Negative_Profit** ($-[\text{Profit}]$) on Size and **Profit** on Color, then adding the **Grand Total**.



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C: Build Sparklines for Trend Analysis

Create a **Sparklines View** by placing **Category** and **Sub-Category** on Rows, **continuous MONTH(Order Date)** on Columns, and **Sales** on Rows to display individual time-series trends. Set independent axis ranges for each row, hide both axes and unnecessary gridlines, add subtle row dividers, and use **Min/Max labels** on **SUM(Sales)** to highlight the minimum and maximum values for each sparkline.

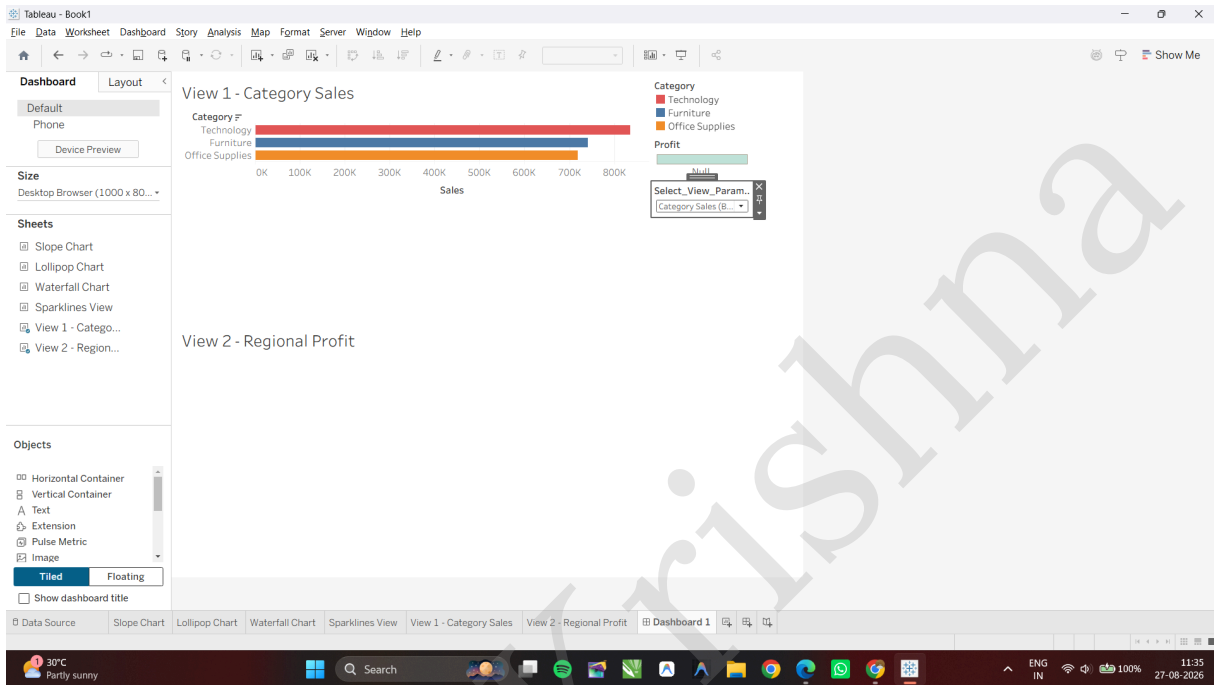


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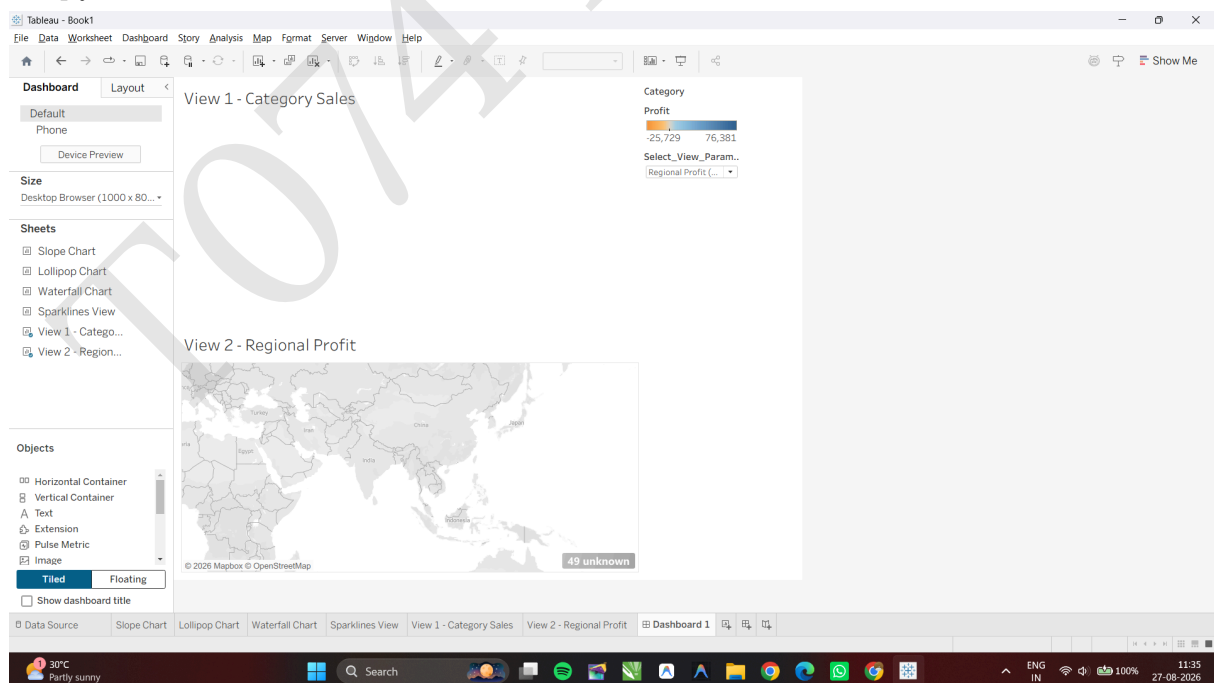
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D: Execute Cross-Database Joins

Create **Select_View_Parameter** and **View_Filter**, build the **Category Sales** and **Regional Profit** views, then use the parameter dropdown to dynamically switch between both sheets on the dashboard.



Select **Regional Profit (Map)**: Sheet 1 collapses completely, and Sheet 2 expands to occupy the container.

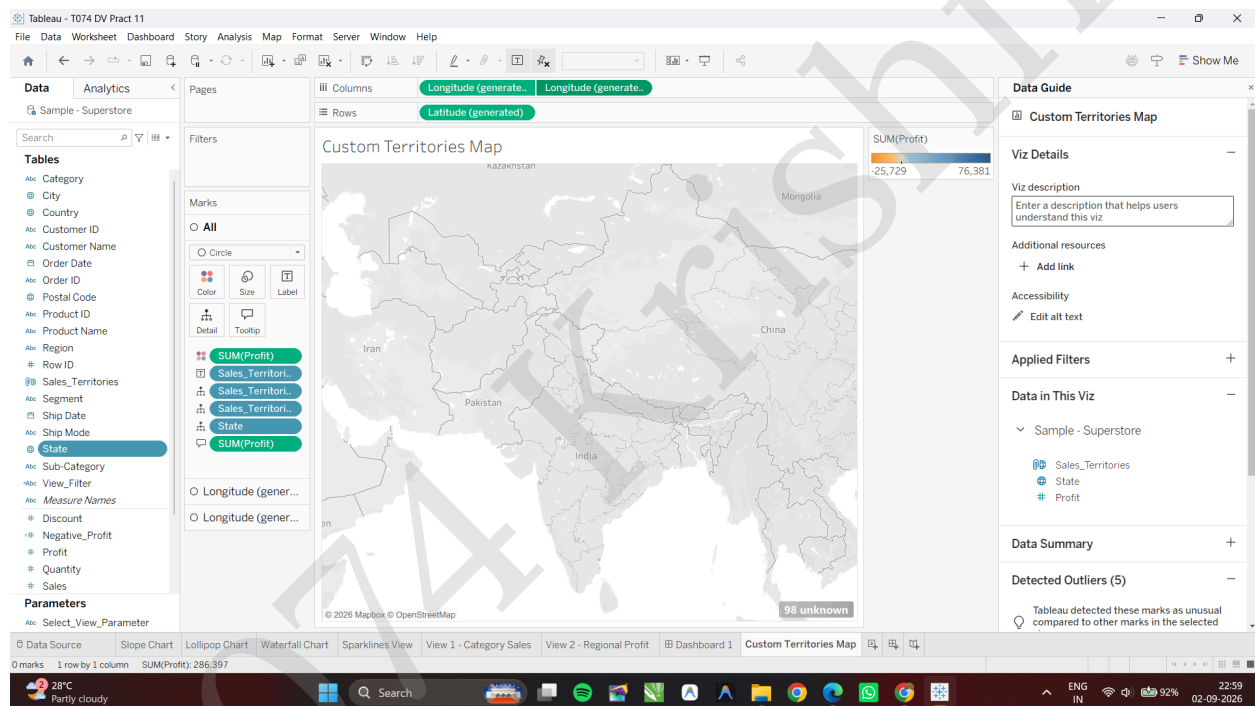


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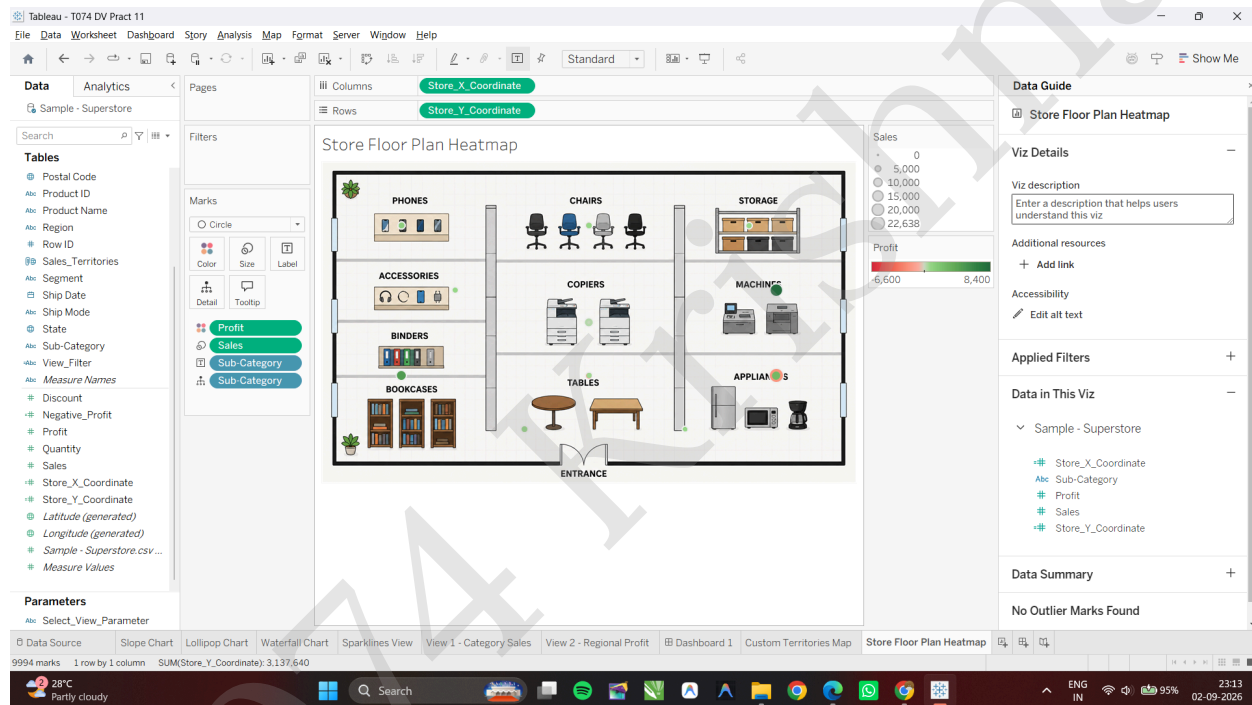
E: Create Custom Geographic Territories

Create a **Custom Territories Map** by grouping the **State** field into four sales territories—**Pacific & Mountain Zone, Midwest & Central Hub, Atlantic & North-East, and Southern Enterprise Territory**—and rename the group field **Sales_Territories**. Build a map using the territory groups with **Sales on Color, Profit on Tooltip, and territory names on Label**, then optionally add a second **State-level Circle layer** using a **Dual Axis** to compare individual states with the custom territory view.



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F: Use Background Images in Dashboards

Create a **Store Floor Plan Heatmap** by making **Store_X_Coordinate** and **Store_Y_Coordinate** calculated fields based on each **Sub-Category**, then add a floor-plan image as a background with the coordinate ranges **X: 0-1000** and **Y: 0-600**. Plot the coordinates using **Circle** marks, add **Sub-Category** to Detail and Label, use **Sales** for Size and **Profit** for Color with a **Red-Green Diverging** palette, then hide the axes and gridlines for a clean floor-plan visualization.



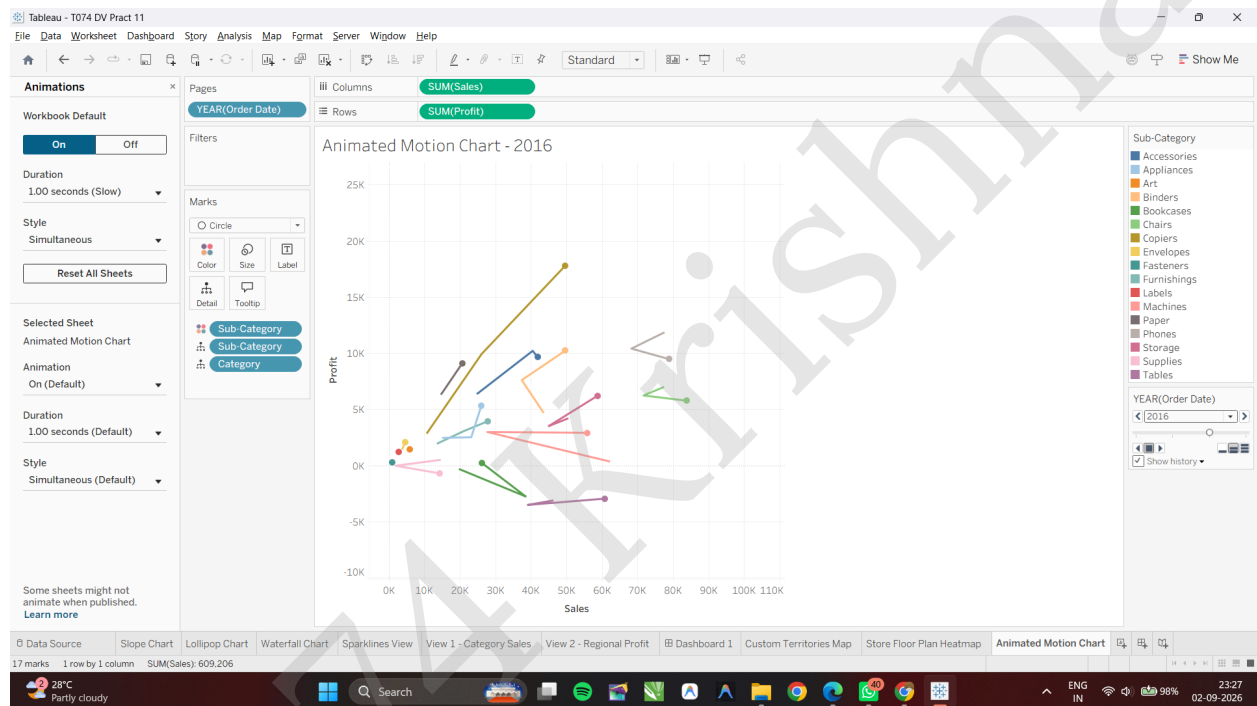
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G: Apply Advanced Data Cleaning Techniques

Task 1: Create Animated Motion Chart

Create an **Animated Motion Chart** by plotting **Sales vs Profit** as circles, using **Sub-Category** for Detail and Color and optionally **Category** for Shape. Add **YEAR(Order Date)** to the **Pages shelf**, enable **Show History** with marks and trails, adjust the playback speed, and turn on **Workbook Animations** to visualize how Sales and Profit change over time.



Task 2: Transparent KPI Card

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Create the **Animated & Transparent Master View** dashboard with a dark background, use **Floating** objects instead of Tiled, and place the **Store Floor Plan Heatmap** (or Custom Territories Map) as the main background. Overlay a **Transparent KPI Card** and the **Animated Motion Chart** as floating elements, resizing and positioning them to create a layered interactive dashboard.

